Product Requirements

UNT AR MAP

Team The Stack
5

Brief problem statement

This should be a 1-2 paragraphs describing what the motivation is for the project. What is the problem to be solved?

New students at the University of North Texas often face difficulties navigating the expansive campus, particularly during their first few weeks. The UNT AR MAP project is designed to address this challenge by providing an Augmented Reality (AR) mobile application that overlays digital navigation markers onto the real-world campus environment using smartphone cameras.

Through this system, students can quickly find buildings, dining halls, departments, and other landmarks. Additionally, the app provides walking directions, estimated time, and context-sensitive tips tailored to the user's location and profile (e.g., freshman, transfer student, faculty). This project aims to reduce stress, enhance campus orientation, and improve the overall UNT student experience. In addition to the mobile AR application, the system also includes a supporting web interface that allows users to view campus maps and project demonstrations outside of the AR environment.

In Sprint 6, the project expands beyond static map visualization by introducing interactive map functionality and web deployment support. Users can now directly click on classrooms and buildings within the map to select navigation destinations, improving usability and reducing reliance on search. Additionally, the system is being adapted for deployment using Unity WebGL, allowing users to access the campus map through a browser without installing the mobile application.

System requirements

What system configuration is needed to run your proposed system (including anything third-party that is needed to run your system).

- Mobile application developed with Unity + AR Foundation
- Support for Android (API 29+) and iOS (iOS 13+)
- Backend services hosted on Firebase (user analytics, feature toggles)
- Compatibility with the latest Chrome and Safari browsers for web-based AR preview
- Integration with UNT campus map APIs and building location services

- Mobile device must include a camera, GPS, and internet access

- Mobile application developed with Unity + AR Foundation
- Support for Android (API 29+) and iOS (iOS 13+)
- Backend services hosted on Firebase (user analytics, feature toggles)
- Compatibility with the latest Chrome and Safari browsers for web-based AR preview
- Integration with UNT campus map APIs and building location services
- Mobile device must include a camera, GPS, and internet access
- Web interface for displaying campus map and project information
- Compatibility with modern browsers (Chrome, Safari, Edge, Firefox)
- Unity WebGL build support for browser deployment
- Hosting support for WebGL builds (e.g., Firebase Hosting or GitHub Pages)
- Mouse and touch input support for interactive map navigation

Users profile

Who is the system intended for? What characteristics should the users have (this can also be a range of things such as reading level, etc.).

- Primary Users: New and transfer UNT students seeking campus navigation - Secondary Users: Faculty, advisors, and current students - Characteristics: - Must own a smartphone with camera + GPS
- Comfortable with mobile apps (no prior AR experience required)
- Willing to allow location + camera permissions
- Reading level: College undergraduate and above

List of Features

Provide a numbered list of features (F1, F2, etc.) that concisely, clearly, and accurately describe that which constitutes your project.

Functional requirements (user stories)

List the Priority as 1 (High Priority - Critical) to 3 (Low Priority – Would be nice if we have time)

F1 – Registration/Login: Users may log in with UNT credentials for personalized suggestions
F2 – AR Navigation: Real-time step-by-step navigation via AR markers
F3 – Location Search: Search for buildings, dining halls, services, and landmarks
F4 – Event Notifications: Alerts about nearby events and info booths
F5 – Accessibility Mode: Text-to-speech and high-contrast overlays for usability
F6 – Heatmap Integration: Real-time display of crowded areas on campus
F7 – Offline Mode: Cached navigation for poor signal areas
F8-Database Integration- Store and retrieve user preferences, building info, and AR marker data from a centralized database

F9 – Web Interface: Web-based platform for displaying campus map and project information
F10– Web Map Rendering: Display campus building layout with interactive, clickable elements for navigation selection
F11- Clickable Classroom Navigation: Users can click or tap classrooms and buildings on the map to select a destination
F12 – Interactive Map System: Map elements respond to user input (click/tap) to trigger navigation and selection logic
F13 – WebGL Deployment: The application can be deployed and accessed through a web browser using Unity WebGL

Functional Requirements (User Stories)

List the Priority as 1 (High Priority - Critical) to 3 (Low Priority – Would be nice if we have time)

No.	User Story Name	Description	Priority
R1	Registration/Login	As a user, I want to log in with UNT credentials so that I can receive personalized recommendations.	2
R2	AR Navigation	As a user, I want to see step-by-step AR directions so I can reach my campus destination.	1
R3	Location Search	As a user, I want to search for any campus building or service so I can find it quickly.	1
R4	Event Notifications	As a user, I want to be notified about nearby events so I don't miss relevant campus activities.	3

R5 Accessibility Mode As a visually impaired user, I want AR text-to-speech so I can use the system effectively. 1

R6 Heatmap Integration As a user, I want to see where the busiest areas are on campus so I can plan my route better. 2

R7 Offline Mode As a user, I want cached navigation so I can still use the app without an internet connection. 2

R8 Database Connectivity As a user, I want the app to store and load campus data and my preferences from a secure database. 1

R9 Web Interface As a user, I want to access a web-based version of the UNT AR Map so I can view campus navigation information and demonstrations through a browser. 2

R10 Web Map Display As a user, I want the campus map to render correctly on the website so I can visually understand building layouts and navigation paths. 1

R11 Clickable Navigation As a user, I want to click on a classroom or building so I can quickly select a destination without typing. 1

R12 Interactive Map As a user, I want the map to respond to clicks so I can explore campus visually and easily select locations. 1

R13 WebGL Access As a user, I want to access the map through a browser so I don't need to install the app. 2

Non-Functional Requirements

Describe any constraints or cross-cutting characteristics of the system in a manner that is clear, specific, and testable. Each requirement should have a unique identifier (e.g. NF1, NF2,..). Only present those that apply to your system. Categories include but are not limited to:

- NF1 – Security: All authentication must use secure UNT SSO (Single Sign-On) with encrypted data storage.
- NF2 – Reliability: The system shall maintain 99% uptime during peak usage (start of semester).
- NF3 – Usability: First-time users should be able to complete a navigation task in under 2 minutes without training.
- NF4 – Cross-Platform Compatibility: The system must run on both iOS and Android devices, and a web interface must be accessible through modern browsers without significant feature differences.
- NF5 – Accuracy: Navigation directions must remain within 3–5 meters of actual building entrances.
- NF6 Database Performance: Database queries should be executed within 2 seconds under normal load conditions.
- NF7 Data Integrity: All stored locations and user data must remain consistent and backed up weekly.
- NF8 – Web Performance: The web interface shall load the campus map and navigation elements within 3 seconds under normal network conditions.
- NF9 – Asset Rendering: All map images and visual assets must render correctly across supported browsers without distortion or missing content.

Sprint 6 Summary

During Sprint 6, the team focused on improving interactivity and expanding platform accessibility. Classrooms and buildings are now clickable within the map interface, allowing users to select navigation destinations directly. Additionally, the system is being adapted for Unity WebGL deployment, enabling access through web browsers. These improvements significantly enhance usability and move the project closer to full cross-platform support.

Sponsor Requirements

I have read and approve the material in this document.

If there is no external sponsor, the TA or instructor will sign it for accuracy/scope.

Print Name Signature Date